

```

// Read and write operations of memory
// Memory size is 64 words of four bits each.
module memory (Enable, ReadWrite, Address, DataIn, DataOut);
  input  Enable, ReadWrite;
  input  [3: 0] DataIn;
  input  [5: 0] Address;
  output [3: 0] DataOut;
  reg [3: 0] DataOut;
  reg [3: 0] Mem [0: 63];                                // 64 x 4 memory

always @ (Enable or ReadWrite or DataIn)
  if (Enable) begin
    if (ReadWrite) DataOut = Mem [Address];                // Read
    else Mem [Address] = DataIn;                            // Write
    else DataOut = 4'bz;
    end                                                    // High impedance state
endmodule

```
